SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	28 October 2026
Team ID	SKILLWALLET-EV-01
Project Title	Electric Vehicle Market Performance & Insights
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Design, develop, and deploy an end-to-end interactive Business Intelligence dashboard analyzing 100,000 global electric vehicles across sales, technical battery performance, pricing, and environmental sustainability.
Scope	Encompasses data cleaning of 37 technical and financial parameters, building 9 modular Tableau visualizations, integrating an interactive executive dashboard, and embedding into a Python Flask web portal.
Problem Statement	
Description	The rapid transition to electric mobility lacks centralized, transparent data benchmarking vehicle range, charging speeds, battery degradation, and total cost of ownership, leading to consumer hesitation and inefficient OEM product planning.
Impact	Empowers prospective EV buyers to make data-backed purchasing choices and enables automotive manufacturers to optimize R&D allocation, battery chemistry selection, and regional pricing strategies.
Proposed Solution	
Approach	Agile BI pipeline leveraging Tableau Desktop for visualization engineering, Tableau Hyper extracts for high-speed in-memory data processing, and Flask for web application delivery.
Key Features	1. Multi-metric global sales & category distribution. 2. Real-world Range vs. Battery Capacity scatter modeling. 3. DC Fast Charging & battery chemistry benchmarking. 4. Price vs. 3-Year Resale Value regression analysis.

	5. Web-embedded responsive dashboard with live interactive filters.

Resource Requirements

Resource Type	Description	Specification/Allocation
Dedicated local workstation		
Computing Resources	System RAM for in-memory data extract and visualization	16 GB DDR4/DDR5
Memory	Local solid state drive for Tableau .hyper extracts and project repo	512 GB NVMe SSD
Storage	Disk space for data, models, and logs	e.g., 1 TB SSD
Python Flask 3.1		
Frameworks	Data manipulation and analytics packages	Pandas, NumPy, python-docx
Libraries	BI & Visualization tools and version control	Tableau Desktop, Tableau Public, VS Code, Git/GitHub
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
100,000 records, 37 attributes, CSV format (~26 MB)		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images